# Sensor Fusion Test Report

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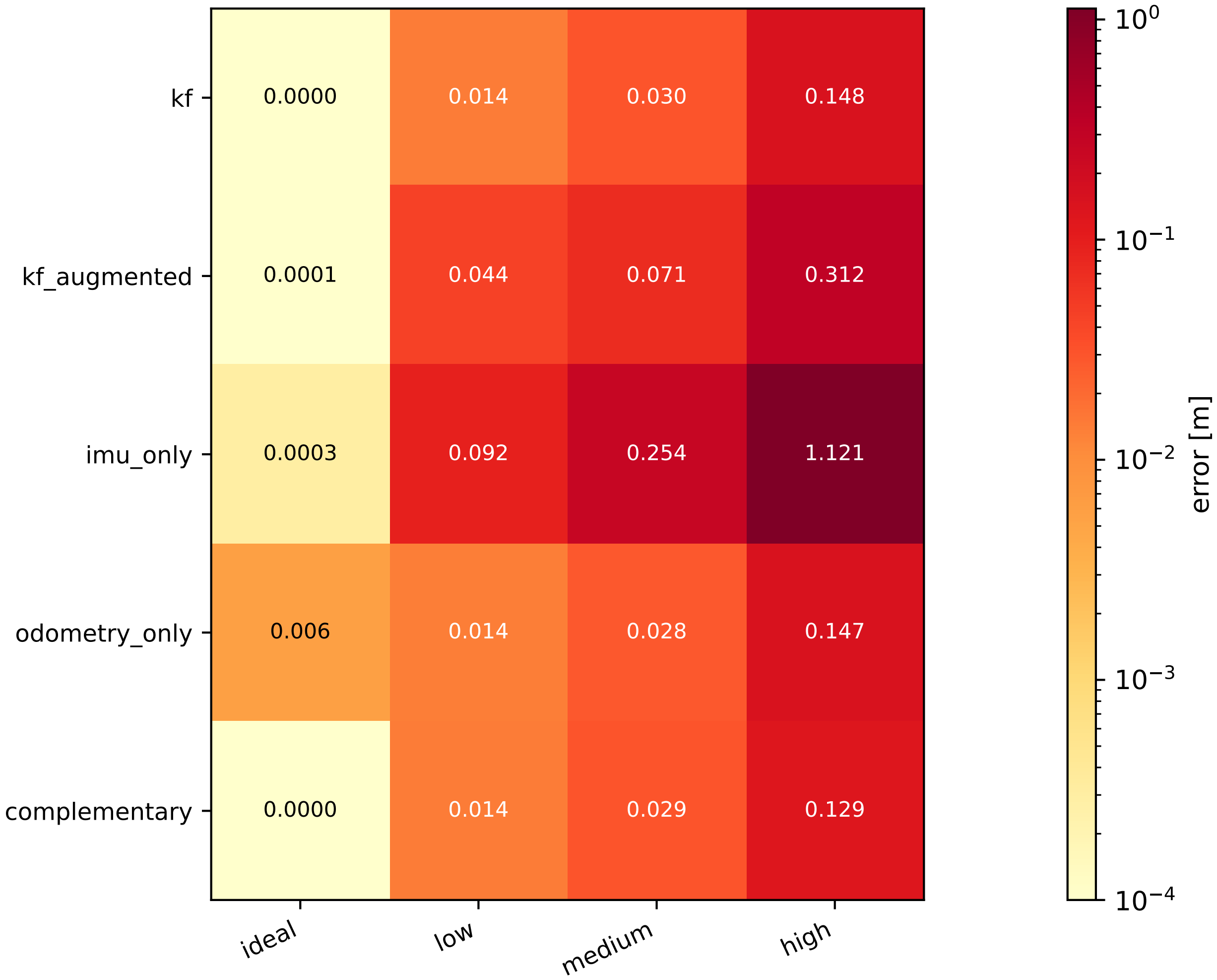
| | |
|---|---|
| **Fusion modes:** | kf, kf_augmented, imu_only, odometry_only, complementary |
| **Trajectories:** | straight, circle, sine, zigzag, figure8, stopgo |
| **Noise levels:** | ideal, low, medium, high |
| **Duration / scenario:** | 8.0 s |
| **Baseline IMU rate:** | 1000 Hz |
| **Baseline wheel rate:** | 50 Hz |
| **IMU rate sweep:** | 250 Hz, 500 Hz, 1000 Hz, 2000 Hz |
| **Wheel rate sweep:** | 10 Hz, 25 Hz, 50 Hz, 100 Hz |
| **Total simulations:** | 90 |

Sections:
1. Noise-level sweep — mode × noise on sine
2. Trajectory sweep — mode × trajectory at MEDIUM noise
3. Frequency sweep — mode × IMU rate / wheel rate
4. Real-time performance — per-step latency + wall-clock miss rate
5. Invariant checks — pass/fail summary

**23/23 invariant checks passed**

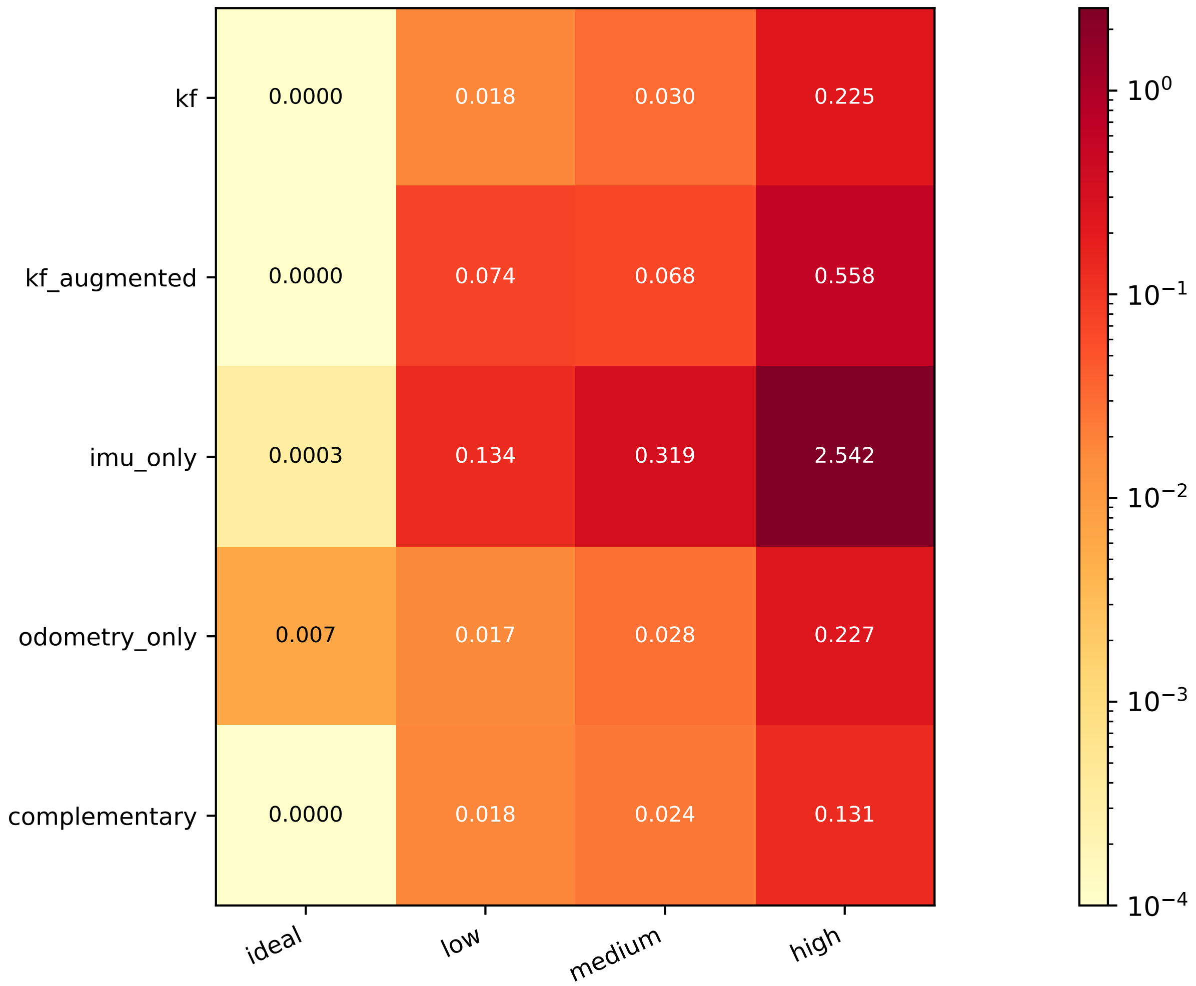
## 1. Position RMS error — mode × noise (sine trajectory)




**Per-mode summary**

```
           kf  best=0.0000 m @ ideal      worst=0.1477 m @ high       ratio=675753.5×
 kf_augmented  best=0.0001 m @ ideal      worst=0.3120 m @ high       ratio=5957.1×
     imu_only  best=0.0003 m @ ideal      worst=1.1212 m @ high       ratio=3801.1×
odometry_only  best=0.0059 m @ ideal      worst=0.1473 m @ high       ratio=25.0×
complementary  best=0.0000 m @ ideal      worst=0.1294 m @ high       ratio=18058.5×
```

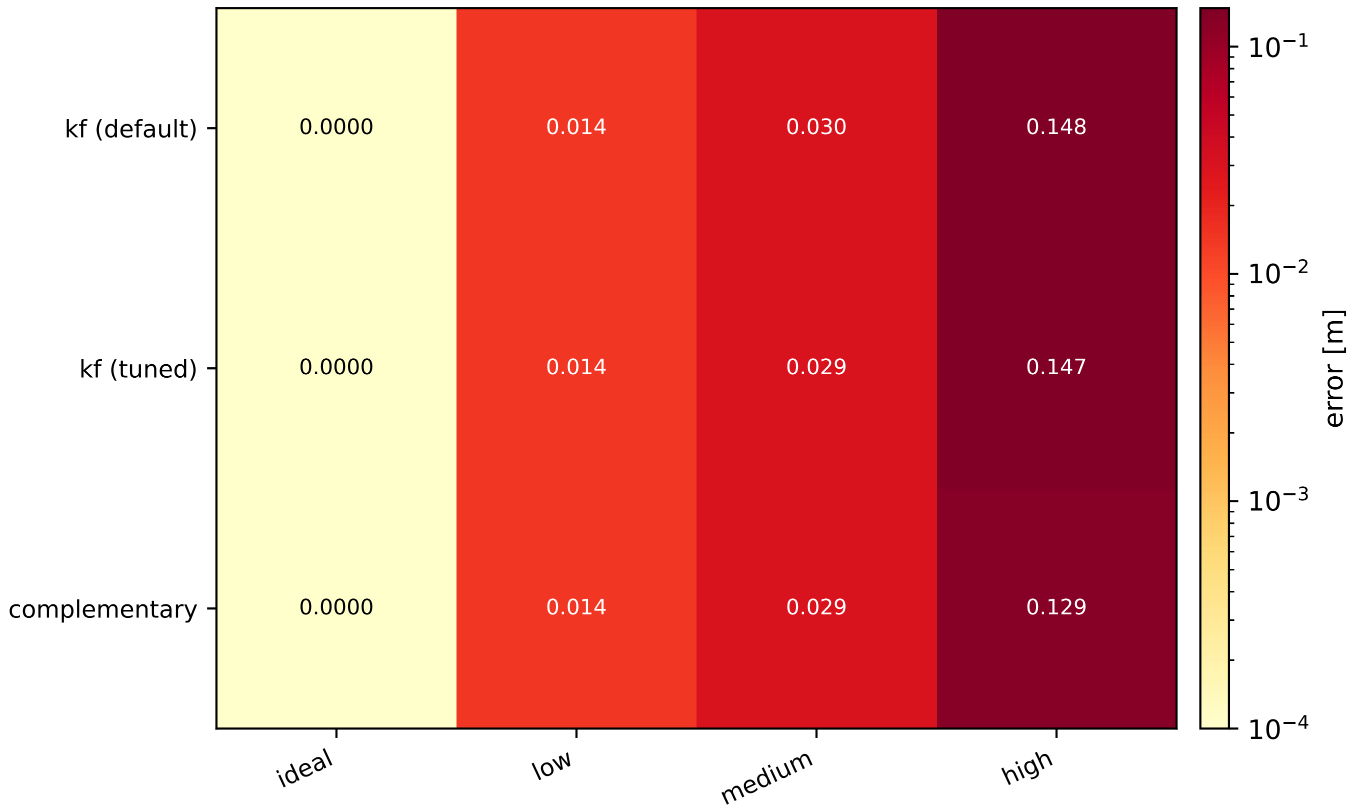
# 1. Final position error — mode × noise (sine trajectory)




**Per-mode summary**

```
           kf  best=0.0000 m @ ideal      worst=0.2250 m @ high      ratio=501895.7×
 kf_augmented  best=0.0000 m @ ideal      worst=0.5583 m @ high      ratio=21207.2×
     imu_only  best=0.0003 m @ ideal      worst=2.5416 m @ high      ratio=7686.8×
odometry_only  best=0.0066 m @ ideal      worst=0.2271 m @ high      ratio=34.3×
complementary  best=0.0000 m @ ideal      worst=0.1315 m @ high      ratio=20064.8×
```

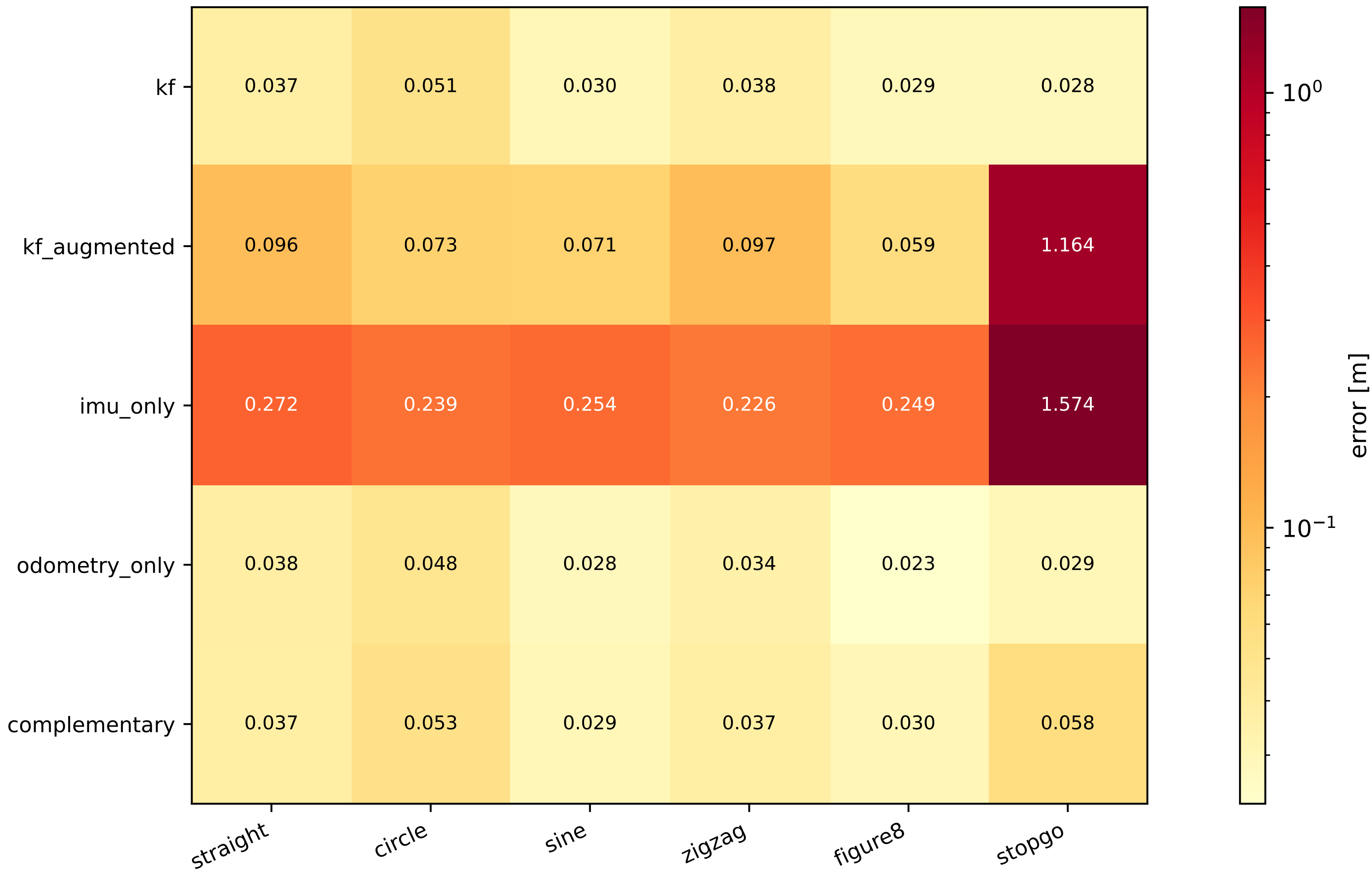
**1b. KF tuning effect — position RMS (sine trajectory)**




**Per-mode summary**

| | | | |
|---|---|---|---|
| kf (default) | best=0.0000 m @ ideal | worst=0.1477 m @ high | ratio=675753.5× |
| kf (tuned) | best=0.0000 m @ ideal | worst=0.1474 m @ high | ratio=369861.4× |
| complementary | best=0.0000 m @ ideal | worst=0.1294 m @ high | ratio=18058.5× |

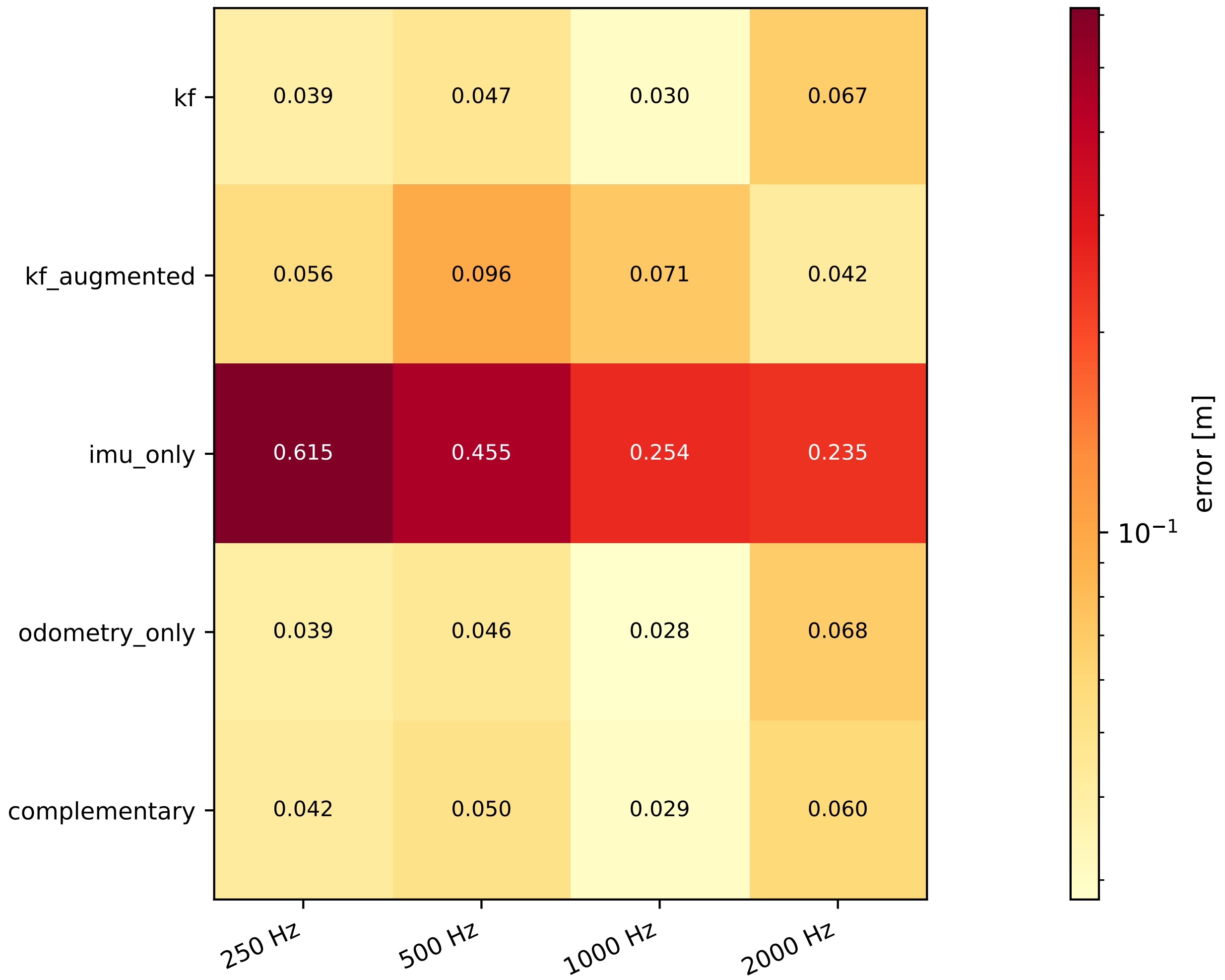
## 2. Position RMS error — mode × trajectory (MEDIUM noise)




**Per-mode summary**

| | | | |
|---|---|---|---|
| kf | best=0.0279 m @ stopgo | worst=0.0512 m @ circle | ratio=1.8× |
| kf_augmented | best=0.0587 m @ figure8 | worst=1.1641 m @ stopgo | ratio=19.8× |
| imu_only | best=0.2257 m @ zigzag | worst=1.5741 m @ stopgo | ratio=7.0× |
| odometry_only | best=0.0232 m @ figure8 | worst=0.0484 m @ circle | ratio=2.1× |
| complementary | best=0.0294 m @ sine | worst=0.0583 m @ stopgo | ratio=2.0× |

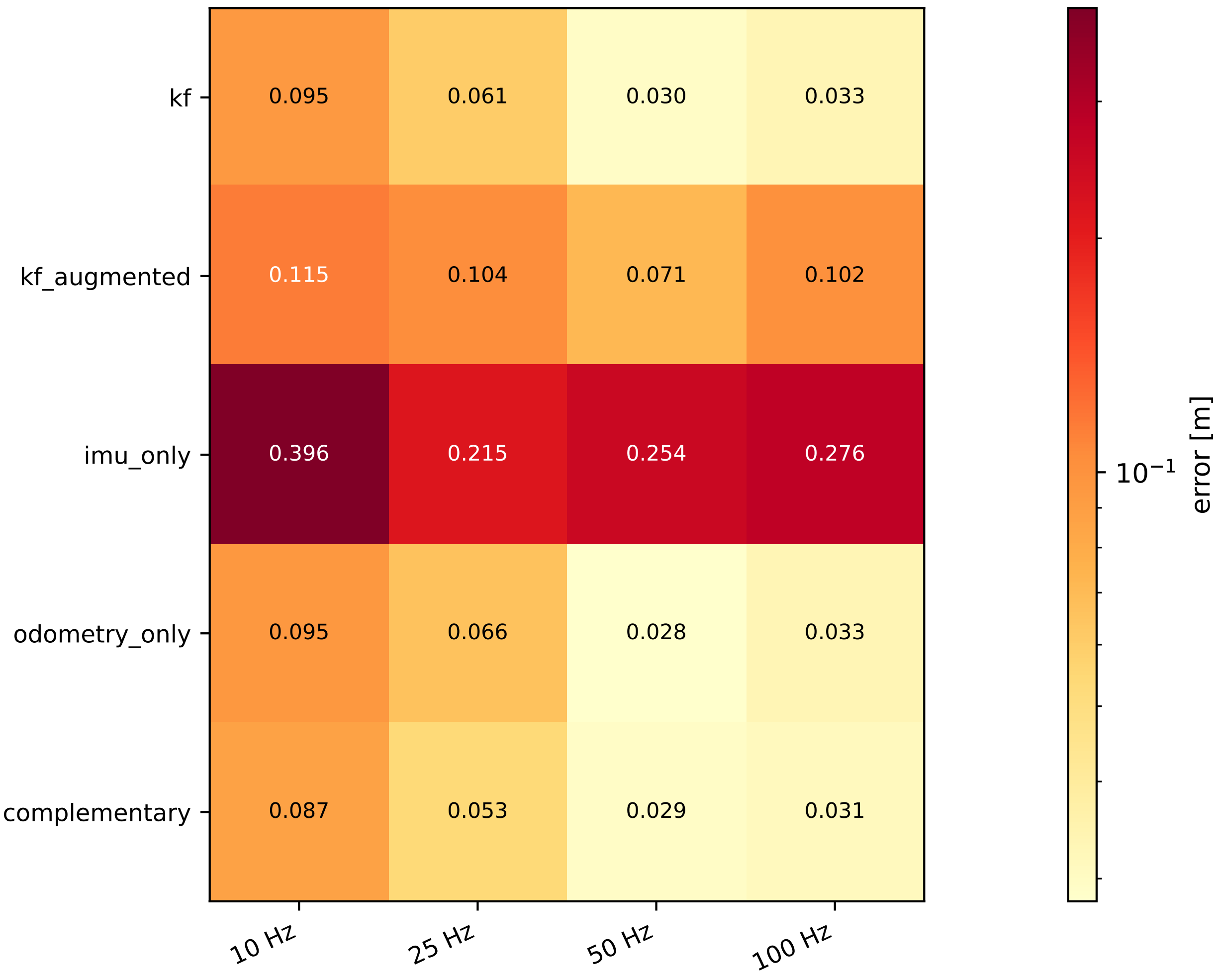
## 3. Position RMS error — mode × IMU rate (sine, MEDIUM)




**Per-mode summary**

| | | | |
|---|---|---|---|
| kf | best=0.0295 m @ 1000 Hz | worst=0.0673 m @ 2000 Hz | ratio=2.3× |
| kf_augmented | best=0.0424 m @ 2000 Hz | worst=0.0961 m @ 500 Hz | ratio=2.3× |
| imu_only | best=0.2350 m @ 2000 Hz | worst=0.6147 m @ 250 Hz | ratio=2.6× |
| odometry_only | best=0.0280 m @ 1000 Hz | worst=0.0681 m @ 2000 Hz | ratio=2.4× |
| complementary | best=0.0294 m @ 1000 Hz | worst=0.0599 m @ 2000 Hz | ratio=2.0× |

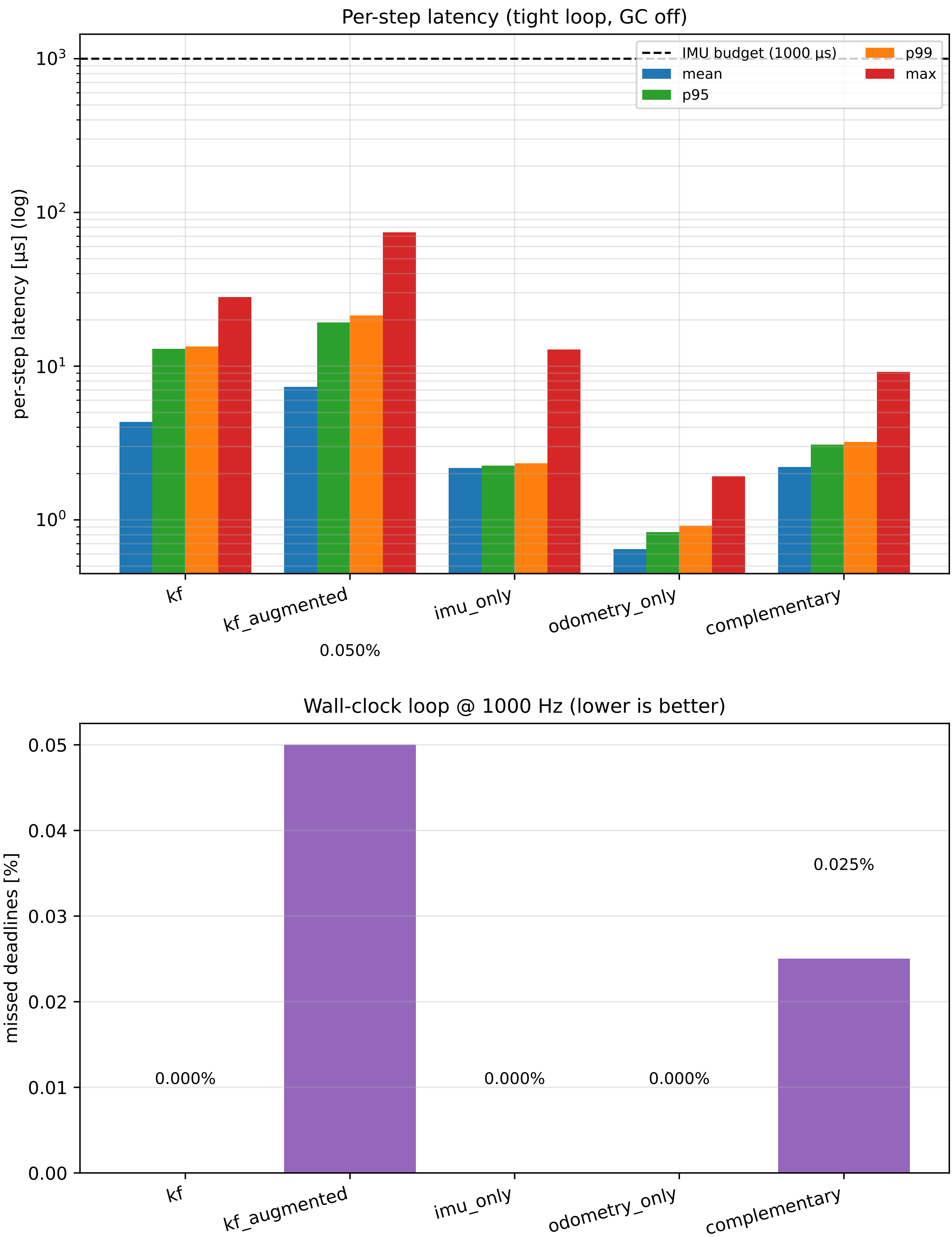
## 3. Position RMS error — mode × wheel-odom rate (sine, MEDIUM)




**Per-mode summary**

| | | | |
|---|---|---|---|
| kf | best=0.0295 m @ 50 Hz | worst=0.0948 m @ 10 Hz | ratio=3.2× |
| kf_augmented | best=0.0713 m @ 50 Hz | worst=0.1154 m @ 10 Hz | ratio=1.6× |
| imu_only | best=0.2150 m @ 25 Hz | worst=0.3956 m @ 10 Hz | ratio=1.8× |
| odometry_only | best=0.0280 m @ 50 Hz | worst=0.0953 m @ 10 Hz | ratio=3.4× |
| complementary | best=0.0294 m @ 50 Hz | worst=0.0867 m @ 10 Hz | ratio=2.9× |

# 4. Real-time performance



# 5. Invariant Checks

23/23 passed

**[PASS]** kf RMS ≤ 5 cm on sine+IDEAL
got 0.0000 m

**[PASS]** kf_augmented RMS ≤ 5 cm on sine+IDEAL
got 0.0001 m

**[PASS]** imu_only RMS ≤ 5 cm on sine+IDEAL
got 0.0003 m

**[PASS]** odometry_only RMS ≤ 5 cm on sine+IDEAL
got 0.0059 m

**[PASS]** complementary RMS ≤ 5 cm on sine+IDEAL
got 0.0000 m

**[PASS]** kf beats imu_only by ≥5× on straight+HIGH
kf=0.1096 m imu=0.8560 m ratio=7.8×

**[PASS]** kf beats imu_only by ≥5× on circle+HIGH
kf=0.0966 m imu=0.6996 m ratio=7.2×

**[PASS]** kf beats imu_only by ≥5× on sine+HIGH
kf=0.1477 m imu=1.1212 m ratio=7.6×

**[PASS]** kf beats imu_only by ≥5× on zigzag+HIGH
kf=0.1145 m imu=0.7610 m ratio=6.6×

**[PASS]** kf beats imu_only by ≥5× on figure8+HIGH
kf=0.0540 m imu=0.7742 m ratio=14.3×

**[PASS]** kf beats imu_only by ≥5× on stopgo+HIGH
kf=0.0461 m imu=0.7982 m ratio=17.3×

**[PASS]** tuned kf ≤ default kf on sine+HIGH
tuned=0.1474 m default=0.1477 m improvement=+0.2%

**[PASS]** kf RMS is monotonic in noise level on sine
0.0000 ≤ 0.0142 ≤ 0.0295 ≤ 0.1477

**[PASS]** kf p99 latency < IMU budget (1000 Hz)
p99=13.46 µs budget=1000 µs

**[PASS]** kf_augmented p99 latency < IMU budget (1000 Hz)
p99=21.33 µs budget=1000 µs

**[PASS]** imu_only p99 latency < IMU budget (1000 Hz)
p99=2.33 µs budget=1000 µs

**[PASS]** odometry_only p99 latency < IMU budget (1000 Hz)
p99=0.92 µs budget=1000 µs

**[PASS]** complementary p99 latency < IMU budget (1000 Hz)
p99=3.21 µs budget=1000 µs

**[PASS]** kf wall-clock miss rate < 1 %
0/4000 (0.000 %) max_overshoot=0.0 µs

**[PASS]** kf_augmented wall-clock miss rate < 1 %
2/4000 (0.050 %) max_overshoot=163.6 µs

**[PASS]** imu_only wall-clock miss rate < 1 %
0/4000 (0.000 %) max_overshoot=0.0 µs

**[PASS]** odometry_only wall-clock miss rate < 1 %
0/4000 (0.000 %) max_overshoot=0.0 µs

(... 2 more checks truncated)